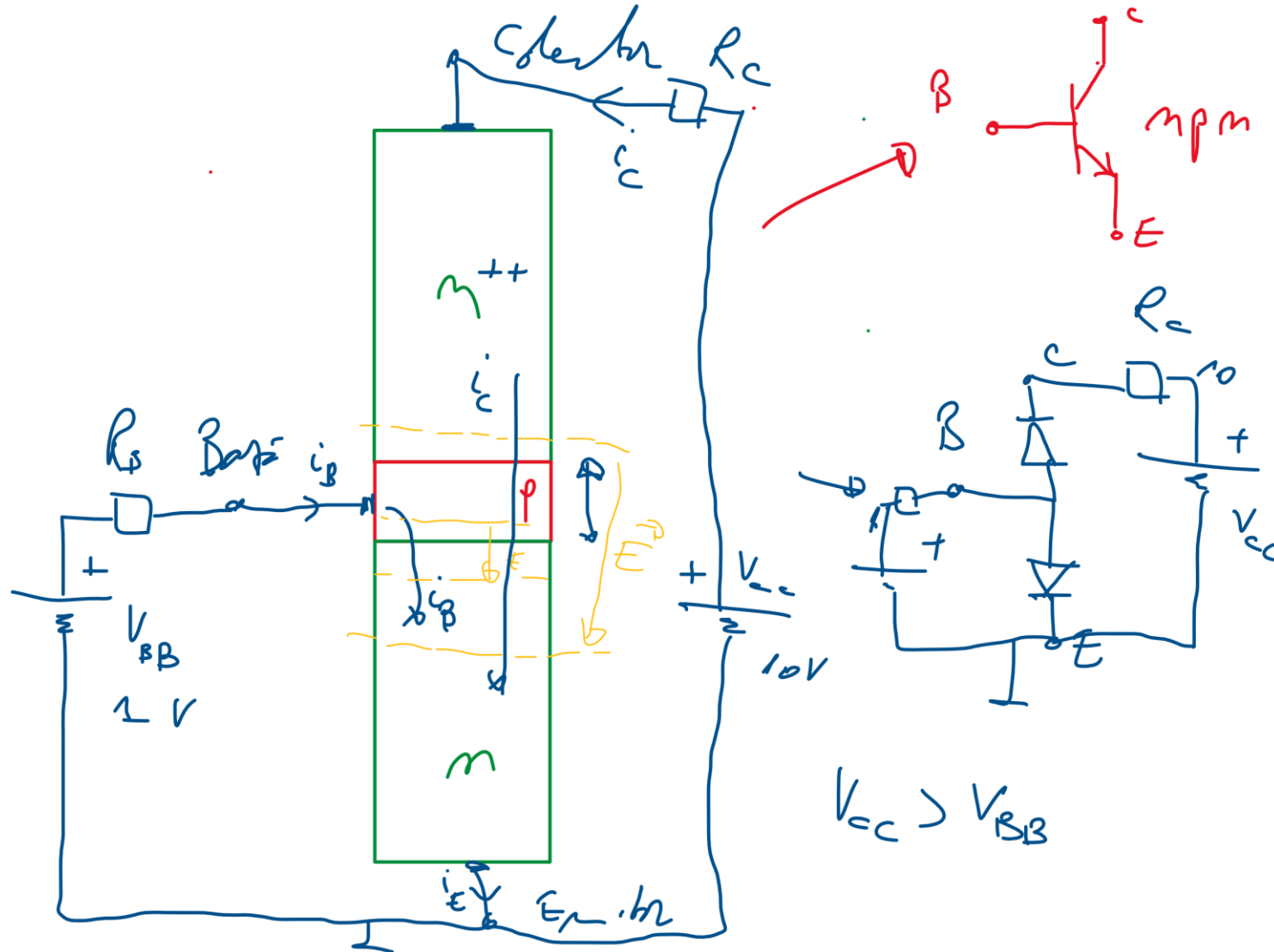


Bazele Circuitelor Electronice

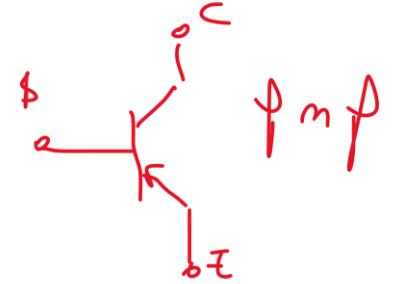
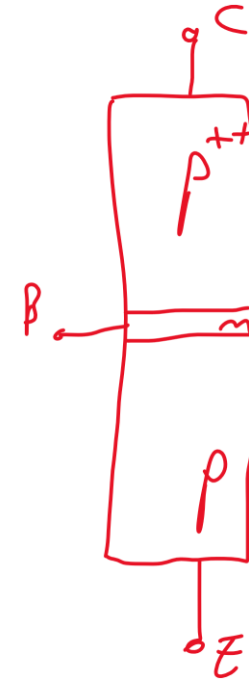
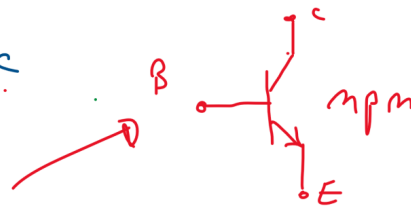
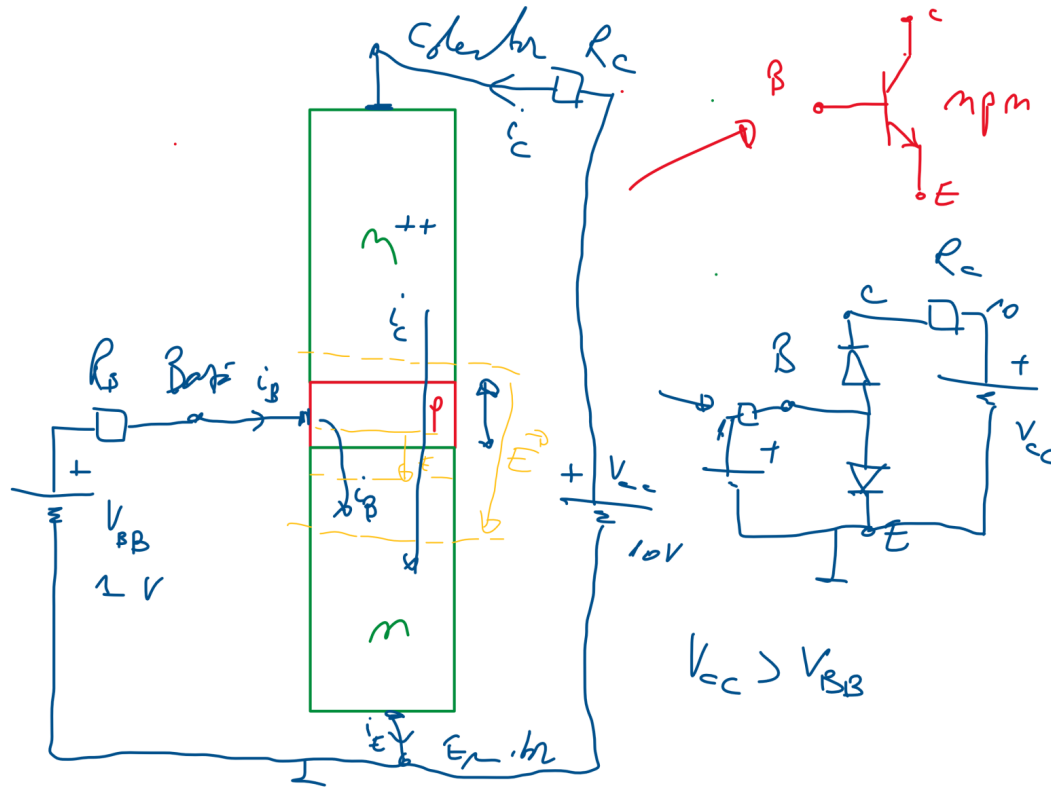
Tranzistorul bipolar



Tranzistorul Bipolar - principiu



Tranzistorul Bipolar - principiu



Tranzistorul Bipolar - principiu

factorul de amplificare în curent $\beta = \frac{I_C}{I_B}$ (1)

$$50 < \beta < 500$$

$$h_{fe} = h_{21} = \beta$$

$$\begin{cases} I_E = I_B + I_C & (2) \\ I_C = \beta I_B & (1) \end{cases}$$

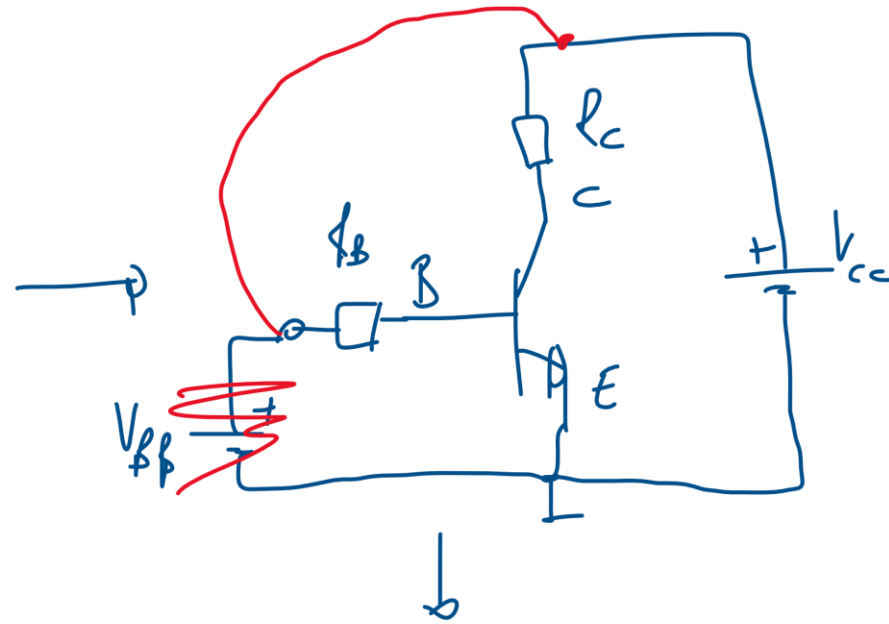
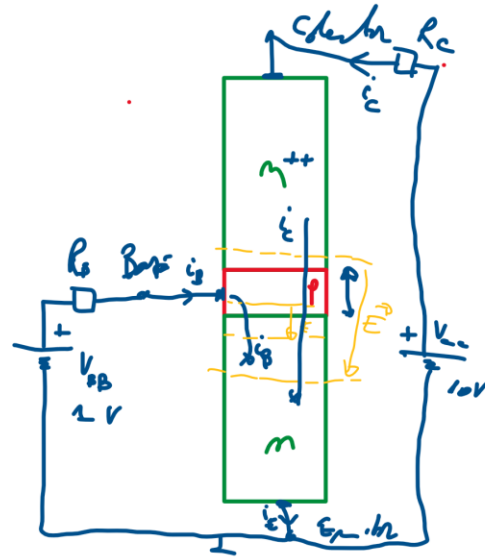
$$\Rightarrow I_E = I_B + \beta I_B = (\beta + 1) I_B$$

$\beta \gg 1$

$$\Rightarrow I_E \approx I_C$$

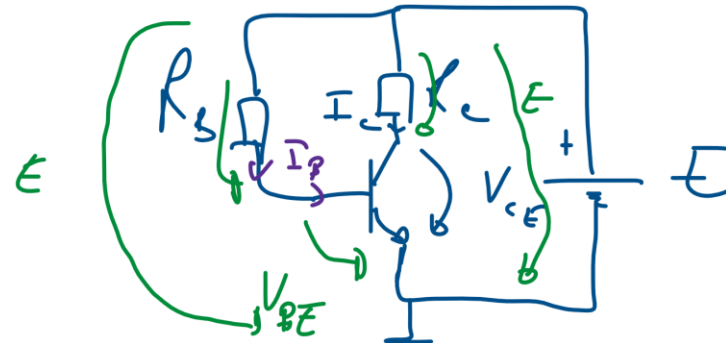


Tranzistorul Bipolar - polarizare



PSF:
 $I_C = ?$ $V_{CE} = ?$

$$V_{BE} = 0.7V$$

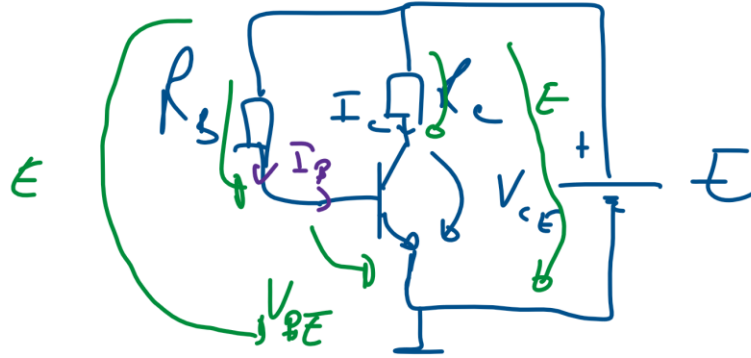


TB – Punctul Static de Funcționare

PSF:

$$I_C = ? \quad V_{CE} = ?$$

$$V_{BE} = 0.7V$$



$$E = R_B \cdot I_B + V_{BE} \Rightarrow I_B = \frac{E - V_{BE}}{R_B}$$

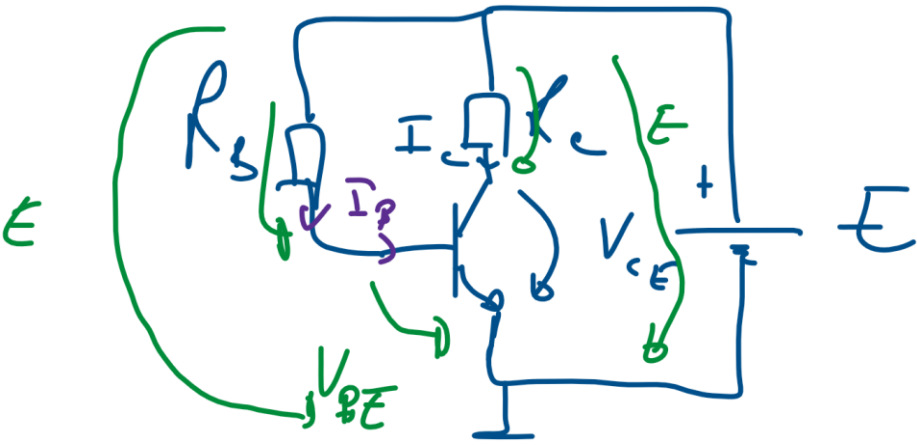


$$I_C = \beta I_B = \beta \frac{E - V_{BE}}{R_B}$$

$$E = R_C \cdot I_C + V_{CE} \Rightarrow \boxed{V_{CE} = E - R_C I_C} \quad (2)$$

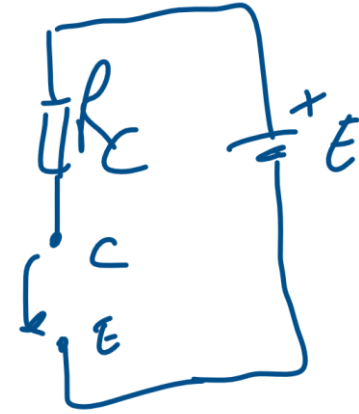


TB – Punctul Static de Funcționare



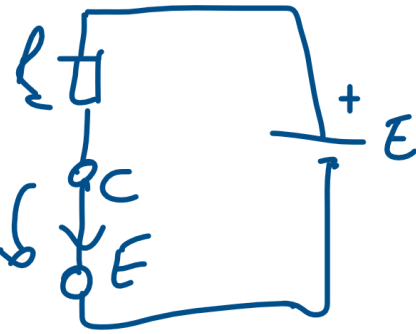
•) T- (G)

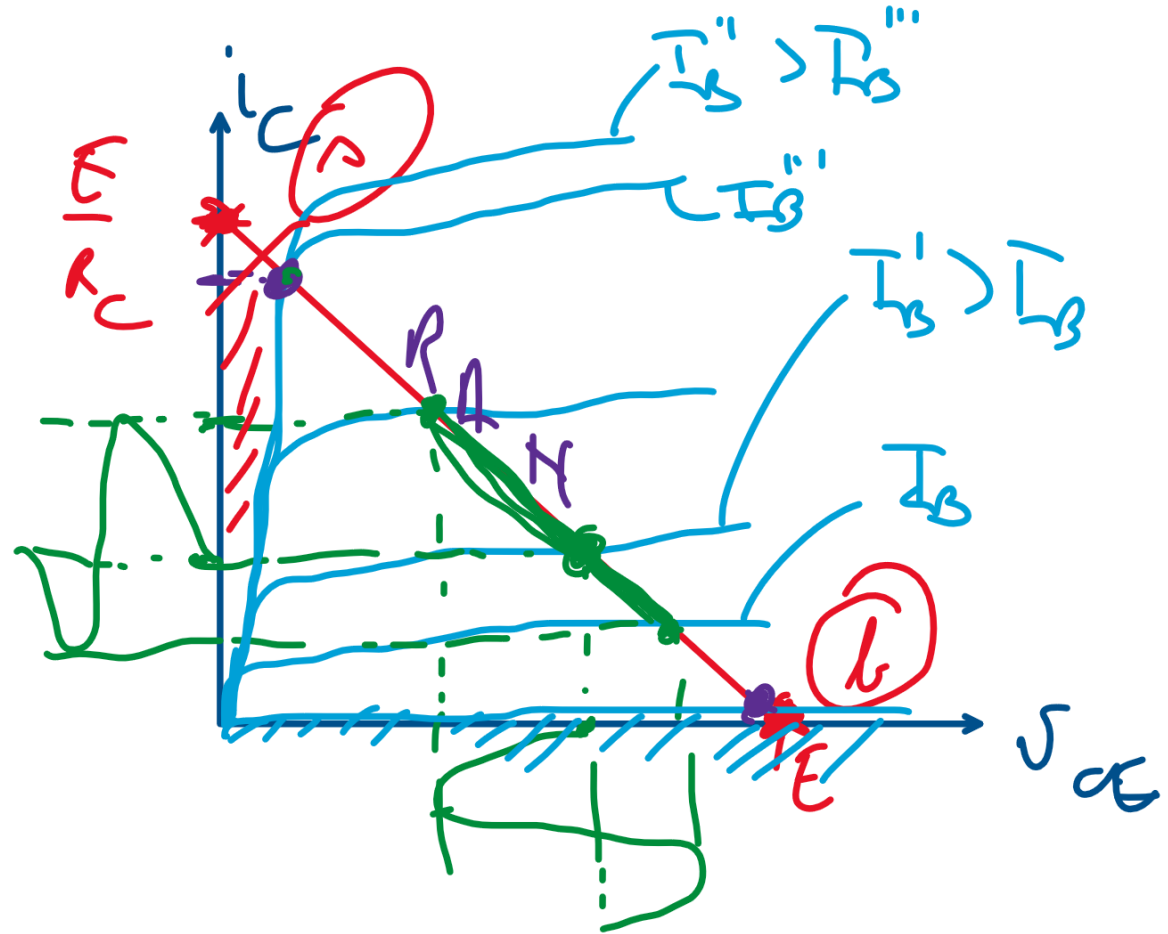
$$\begin{cases} I_C = 0 \\ V_{CE} = E \end{cases}$$



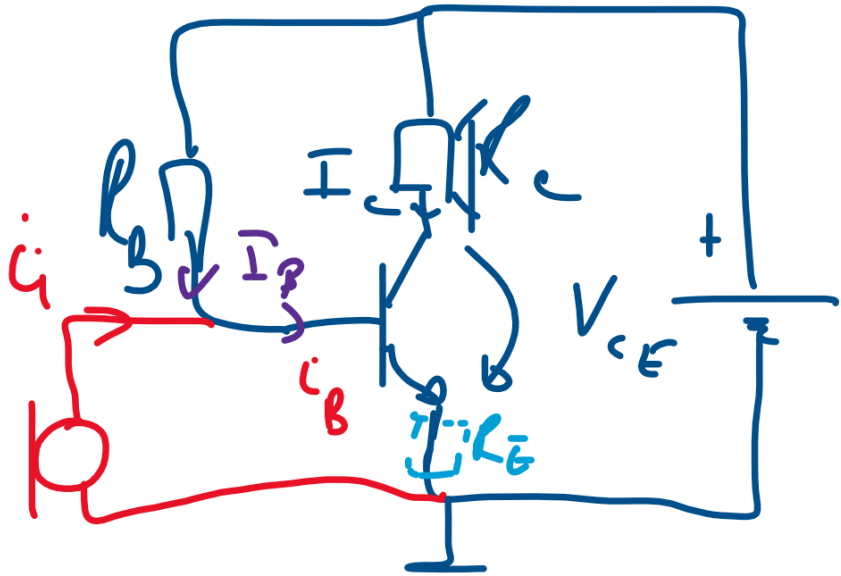
•) T- (D)

$$\begin{cases} V_{CE} = 0 \\ I_{C_{max}} = \frac{E}{R_C} \end{cases}$$

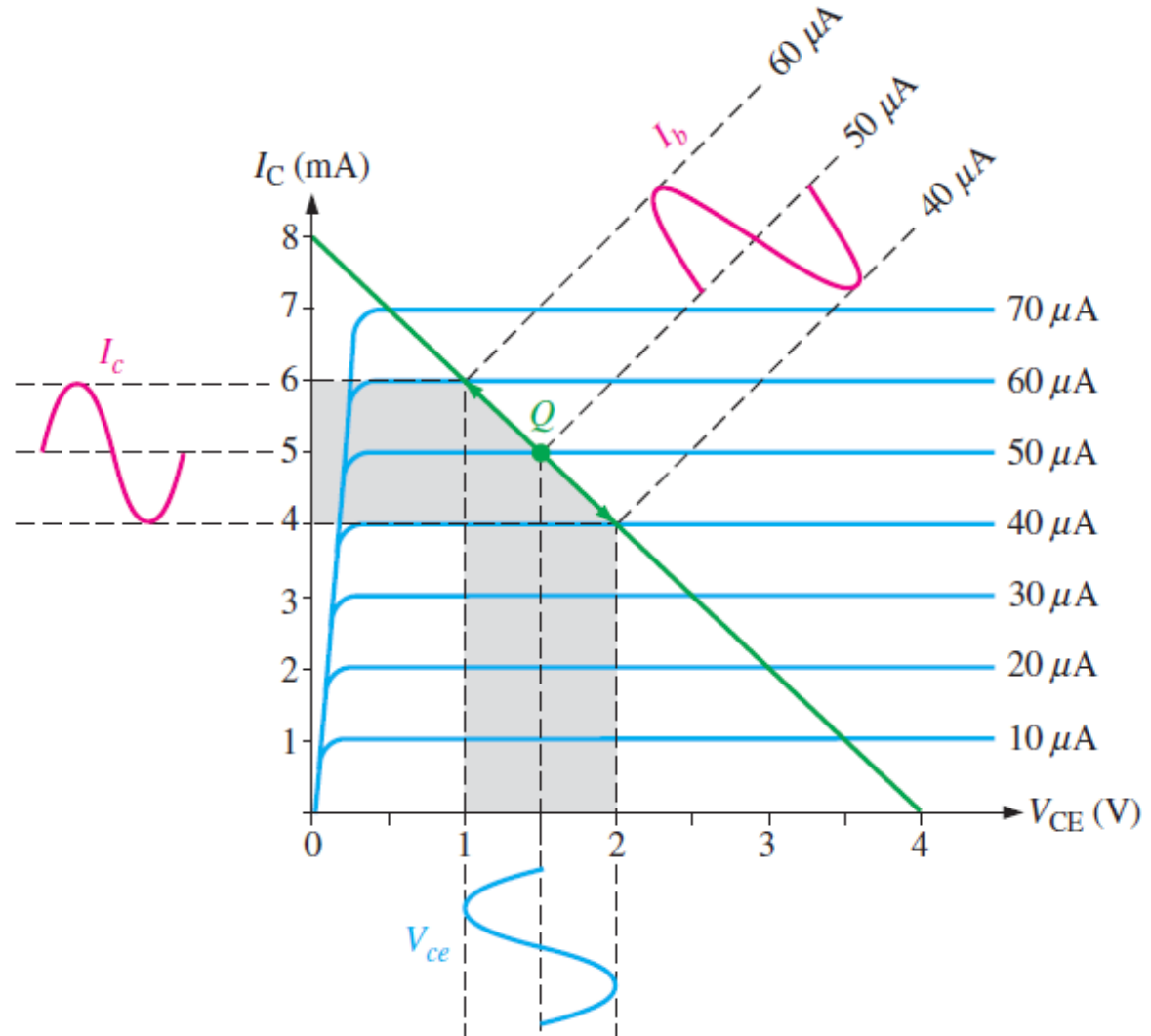




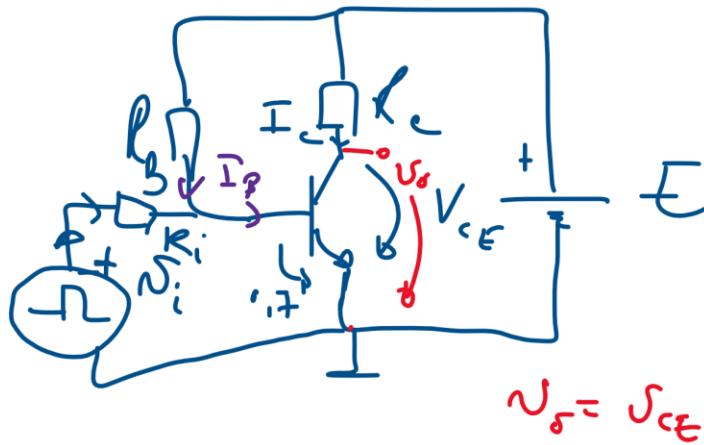
TB – Punctul Static de Funcționare



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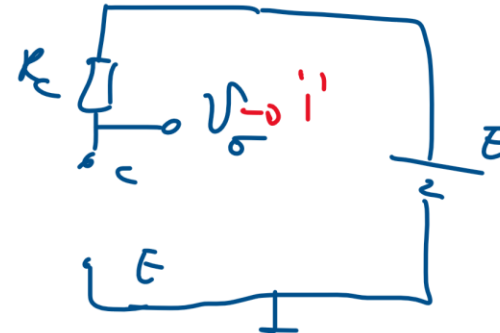
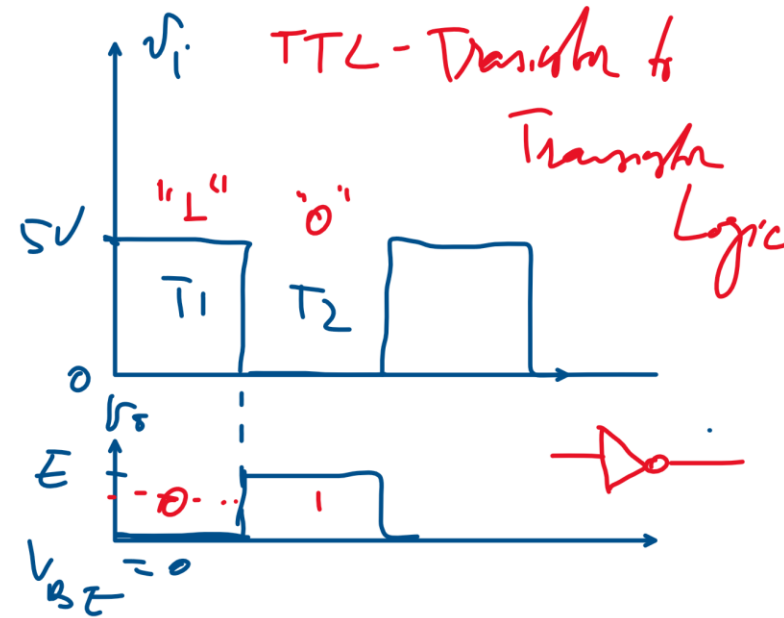
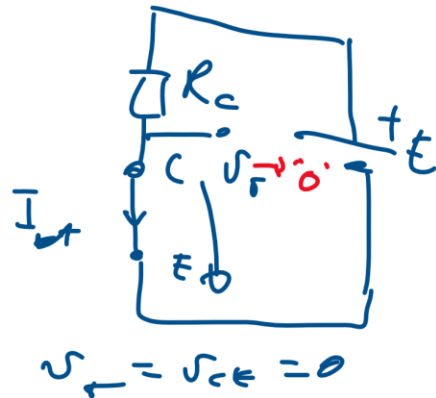


TB – regimul de comutație



T1: $V_{BE} \approx 0.7V$

T2: $V_{BE} = 0$



$v_o = v_{CE} = E$



TB – regimul de comutație

$$I_c = \beta I_B$$

$$I_{c_{max}} \leq \beta I_B = \beta \frac{E - V_{BE}}{R_B}$$

$$\frac{E}{R_c} \leq \beta \frac{E - V_{BE}}{R_B} \Rightarrow$$

$$V_{BE} \ll E$$

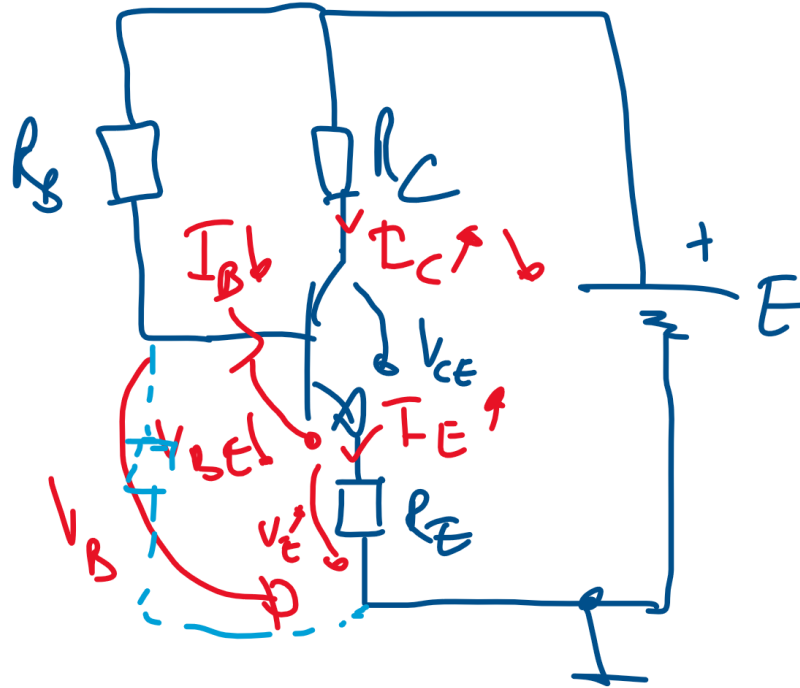
$$\frac{1}{R_c} \leq \frac{\beta}{R_B}$$

$$\boxed{R_B \leq \beta R_c}$$

cond. de
comutație



Circuit polarizare cu rezistență în emitor

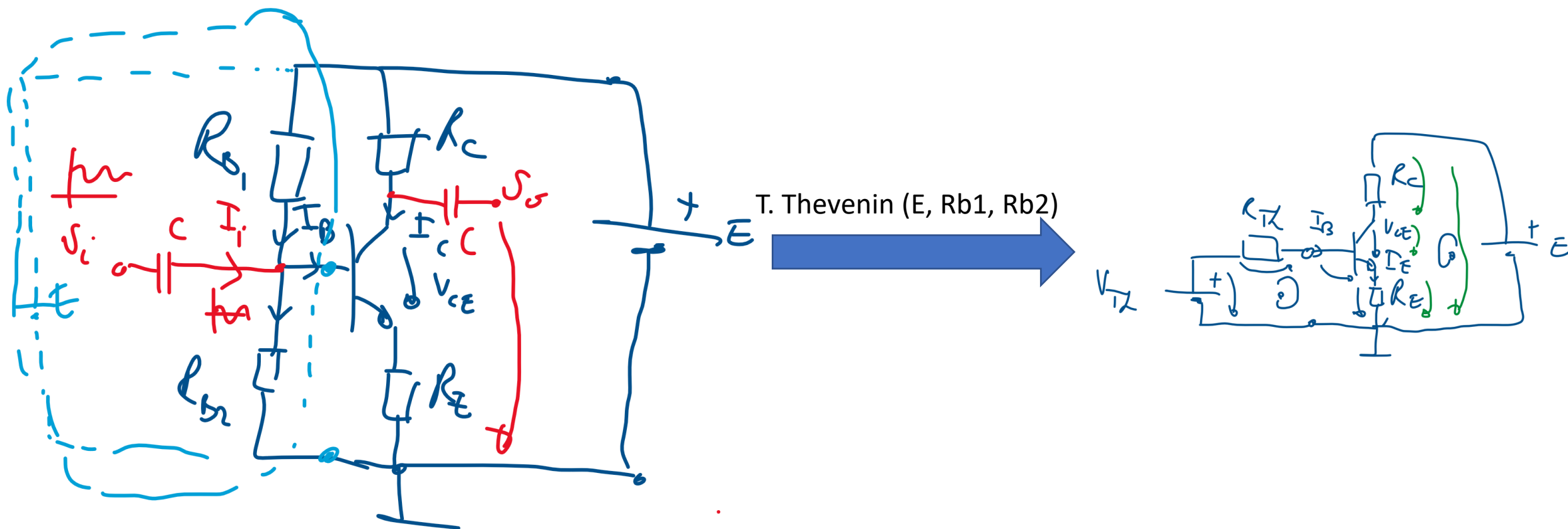


$$V_{BE} = V_B - V_E = V_B - R_E \times I_E$$

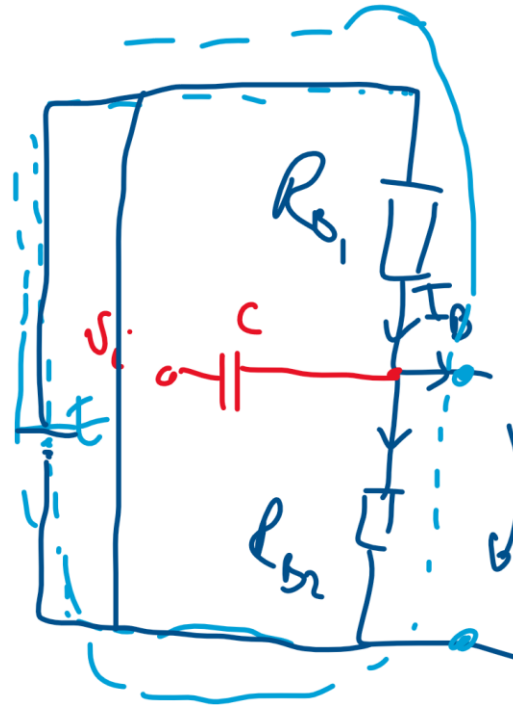
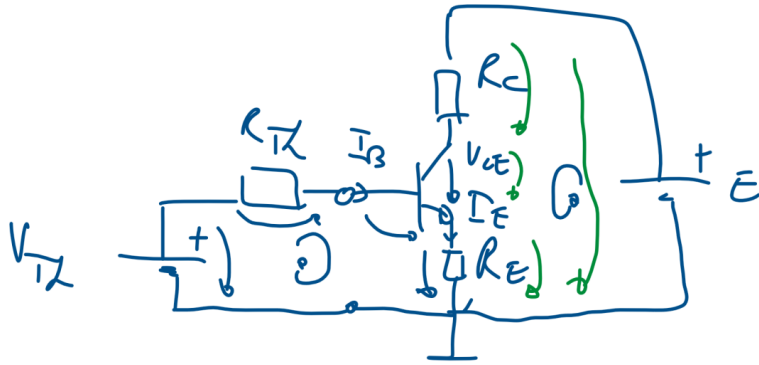
The equation is written in red ink. A bracket under V_E points to the term $R_E \times I_E$, which is enclosed in a dashed red circle. A red arrow points from the circle to the right.



Circuit polarizare cu divizor în bază



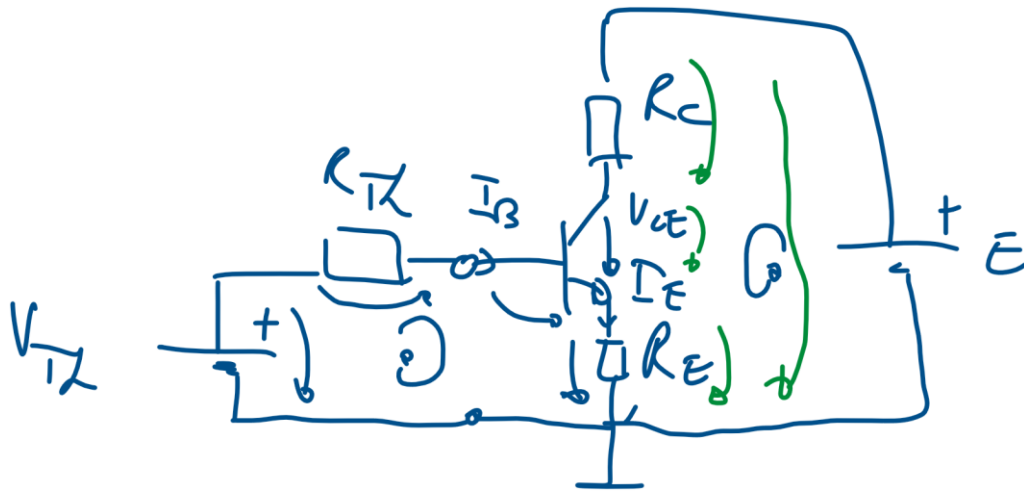
Circuit polarizare cu divizor în bază



$$V_{T2} = \frac{R_{B2}}{R_{B1} + R_{B2}} E \quad ; \quad R_{T2} = R_{B1} \parallel R_{B2}$$



Circuit polarizare cu divizor în bază



$$V_{T2} = R_{T1} \cdot I_B + V_{BE} + R_E \cdot I_E$$

$$I_E = I_B + I_C = I_B + \beta I_B = (\beta + 1) I_B$$

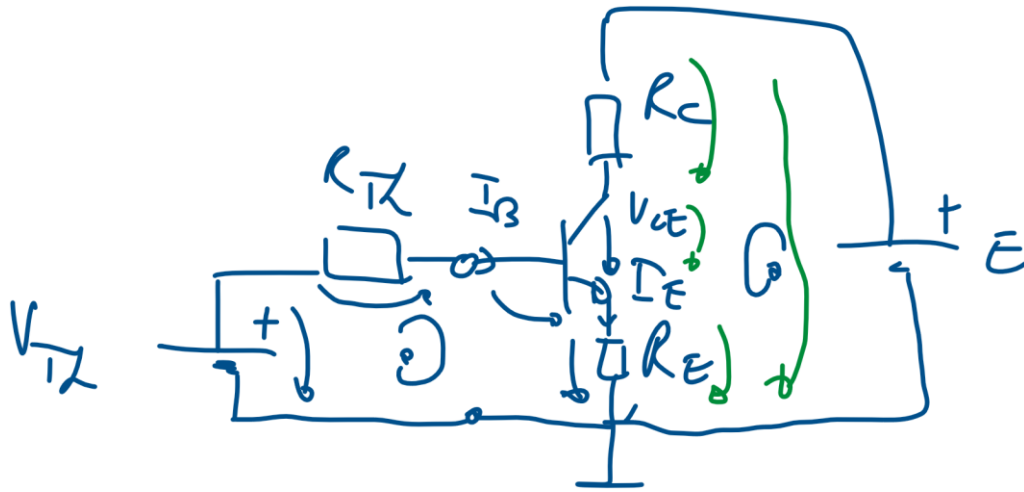
$$V_{T2} = R_{T1} I_B + V_{BE} + (\beta + 1) I_B \cdot R_E$$

$$I_B = \frac{V_{T2} - V_{BE}}{R_{T1} + (\beta + 1) R_E} \quad I_C = \beta I_B$$



Circuit polarizare cu divizor în bază

$$E = R_c I_c + V_{CE} + R_E I_E$$



$$\left. \begin{aligned} I_E &= (\beta + 1) I_B \\ I_B &= \frac{I_c}{\beta} \end{aligned} \right\} \Rightarrow I_E = \frac{\beta + 1}{\beta} I_c$$

$$E = R_c I_c + V_{CE} + R_E \frac{\beta + 1}{\beta} I_c$$

$$V_{CE} = E - R_c I_c - R_E \frac{\beta + 1}{\beta} I_c$$





2N2222 pinout

1. Emitter
2. Base
3. Collector

